

Data Handling Summary

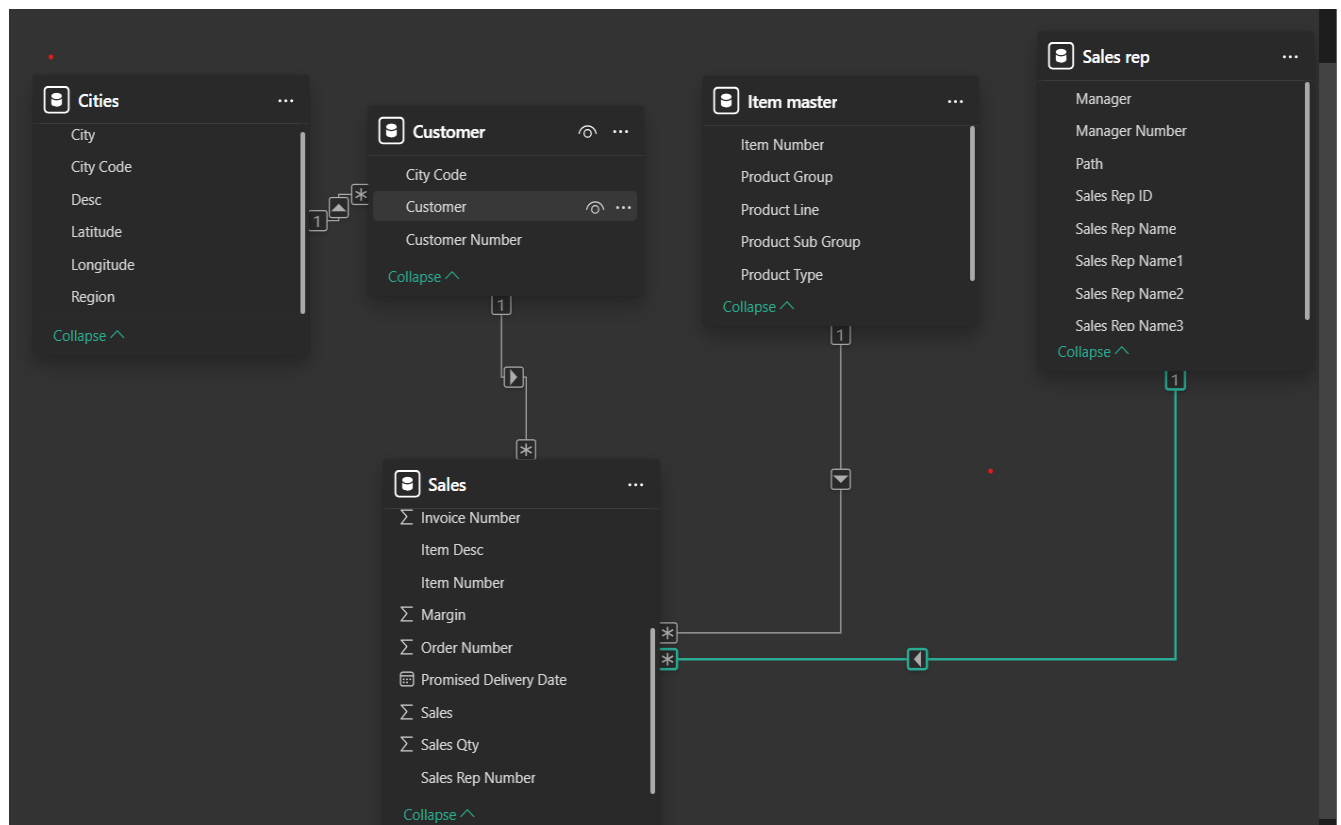
Here are the steps I took in my data handling and visualization process.

Step 1: Importing and Cleaning the Data

- I Imported 5 datasets: **Cities.xlsx**, **Customers.xlsx**, **Item master.xlsx**, **Sales rep.csv**, and **Sales.xlsx** into Power BI.
- I make sure there are no duplicates in all of my data sets. Note here that most of my data cleaning process is taken care of in my excel sheet rather than power BI. Even though the data is pretty much clean, I took some actions in like Splitting the Sales rep data in Excel to create separate columns for **Sales Rep ID**, **Sales Rep Name**, and **Manager**.
- I also ensured Date columns in the Sales table were set to **Date type**.

Step 2: Data Modeling (Creating Relationships)

Created the following relationships in Model View:



Step 3: Creating Calculated Columns

I Created 4 columns in the Sales table using the Date field (actual sale date):

year = YEAR(Sales[Date])

Quarter = QUARTER(Sales[Date])

Month = FORMAT(Sales[Date], "mmmm")

Month Number = MONTH(Sales[Date])

- I Used Month Number to sort Month chronologically (Column Tools → Sort by Column).

Step 4: Creating a Measure

- I created a Gross Margin measure to analyze profitability across products, regions, and customers.

Gross Margin (%) = DIVIDE(SUM(Sales[Sales]) - SUM(Sales[Margin]), SUM(Sales[Sales]))

- I Also created a second measure analyze the average sales per invoice.

Average sales per Invoice =

DIVIDE(SUM(Sales[Sales]), DISTINCTCOUNT(Sales[Invoice Number]))

Step 5: Designing Dashboards and Visualizations

- I Created four separate dashboards to support business questions:

1. High-Level Dashboard

2. Product Details Dashboard

3. Customer Details Dashboard

4. Customer Location Dashboard

Note:

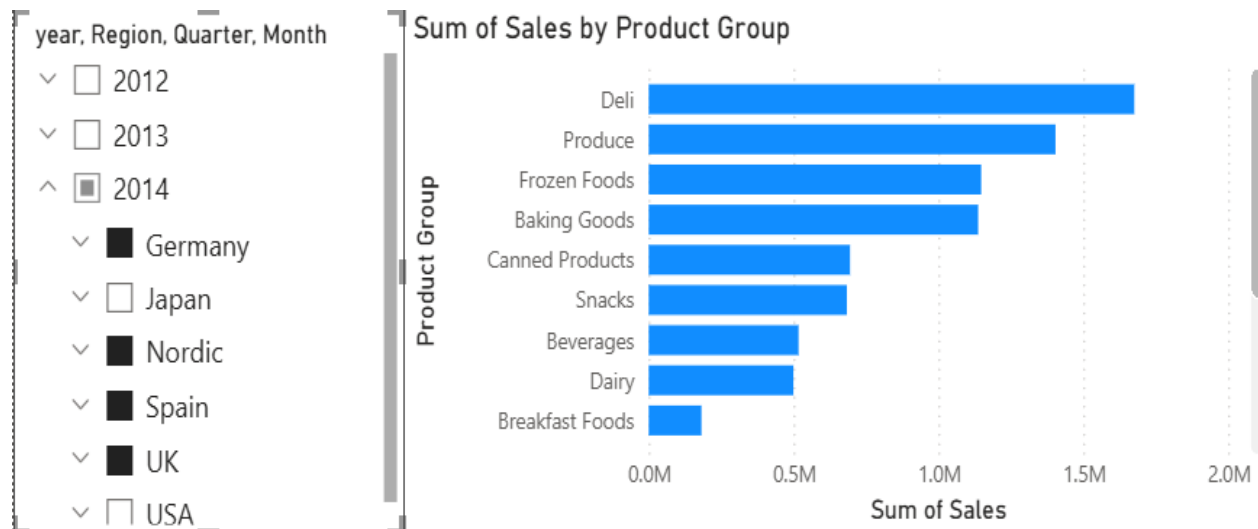
I Use **SUM** for all numeric aggregations (Sales, Margin, Sales Qty) unless **AVERAGE** was explicitly required using a custom measure(Average sale per invoice).

Answer to analysis questions

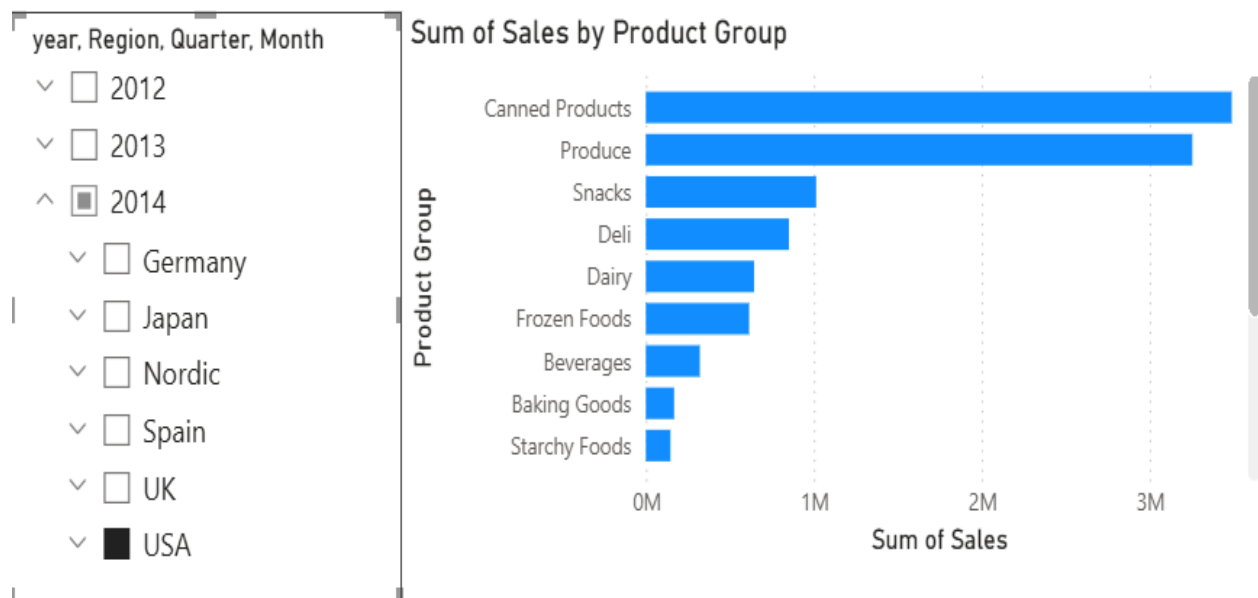
Question 1:

For both the U.S. and for Europe, in 2014 YTD (year to date), what is the top-selling product group in each region?

For Europe(Germany, Nordic, Spain, UK)



For USA



In 2014 YTD, the top-selling product group in the U.S. is Canned Products, while in Europe it is Deli.

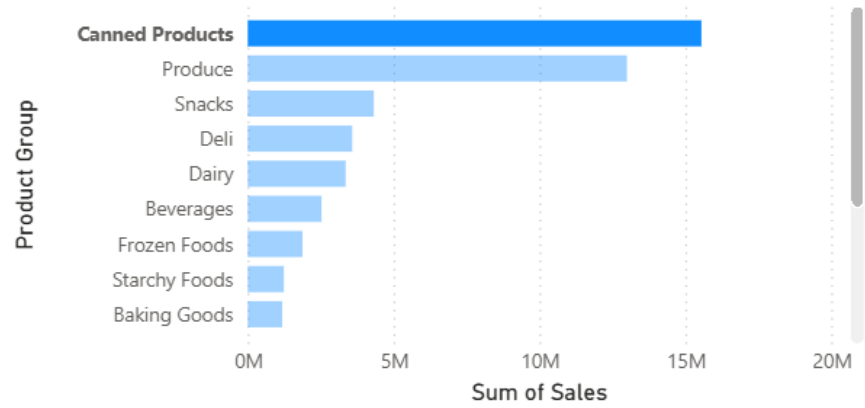
Question 2:

What are the sales trends for this top product group for the U.S. and Europe from 2012 through 2014?

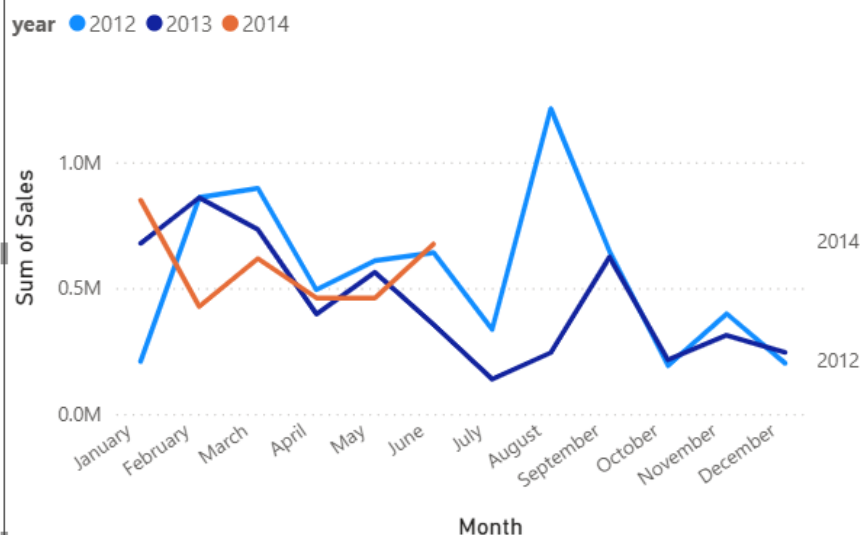
For USA

- ^ 2012
 - ✓ Germany
 - ✓ Japan
 - ✓ Nordic
 - ✓ Spain
 - ✓ UK
 - ✓ USA
- ^ 2013
 - ✓ Germany
 - ✓ Japan
 - ✓ Nordic
 - ✓ Spain
 - ✓ UK
 - ✓ USA
- ^ 2014
 - ✓ Germany
 - ✓ Japan
 - ✓ Nordic
 - ✓ Spain
 - ✓ UK
 - ✓ USA

Sum of Sales by Product Group



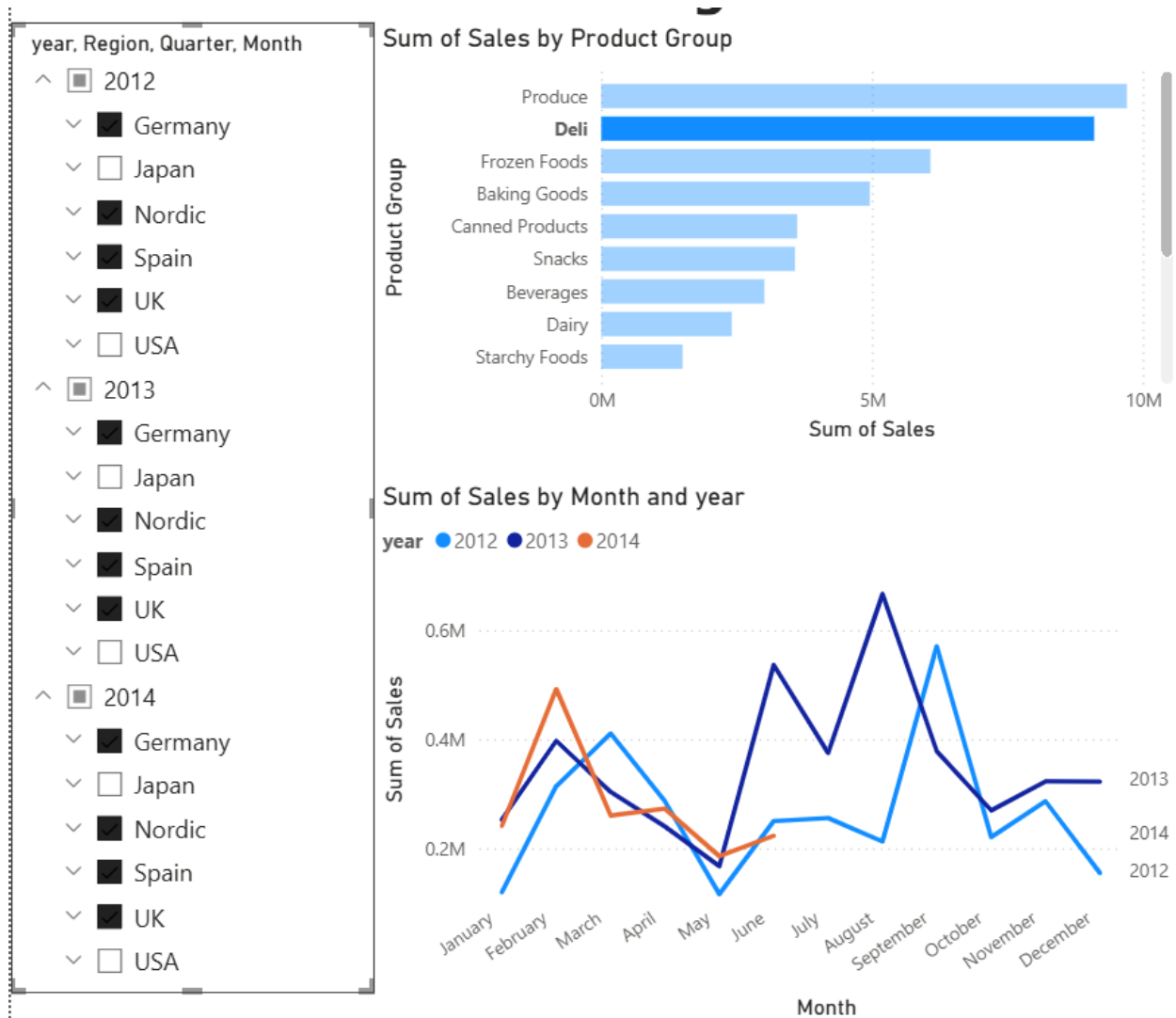
Sum of Sales by Month and year



For US, the top-selling product group in 2014 was Canned Products. Looking at trends from 2012 to 2014, 2012 had the highest overall sales, with a strong peak in August and consistent performance in February and March.

2013 showed similar seasonality but with lower overall sales, and a modest rise in September. 2014 had the lowest performance, with an early spike in January, followed by relatively flat sales throughout the rest of the year.

For Europe



For, the top-selling product group in 2014 was Deli.

From 2012 to 2014, 2013 had the strongest sales, peaking in August.

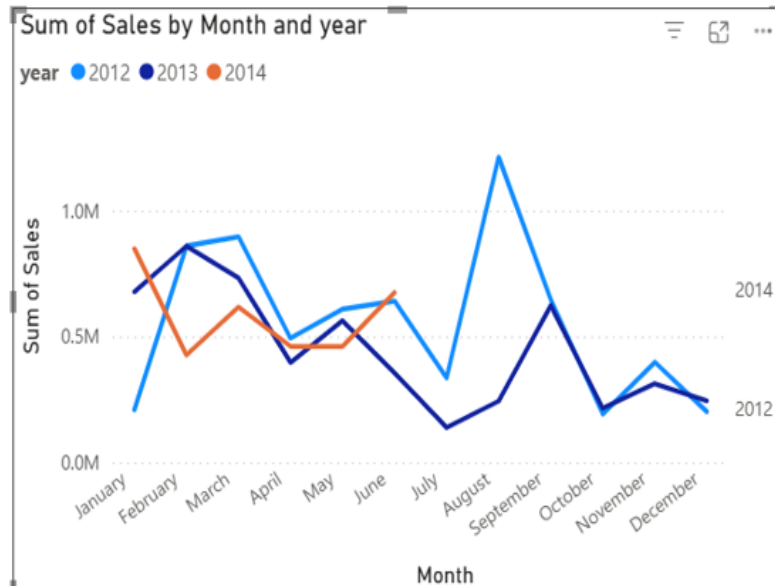
2012 also showed a seasonal pattern, with lower overall sales.

2014 had a good start in February, but remained flat throughout the rest of the year, making it the weakest of the three years in total sales for Deli.

Question 3:

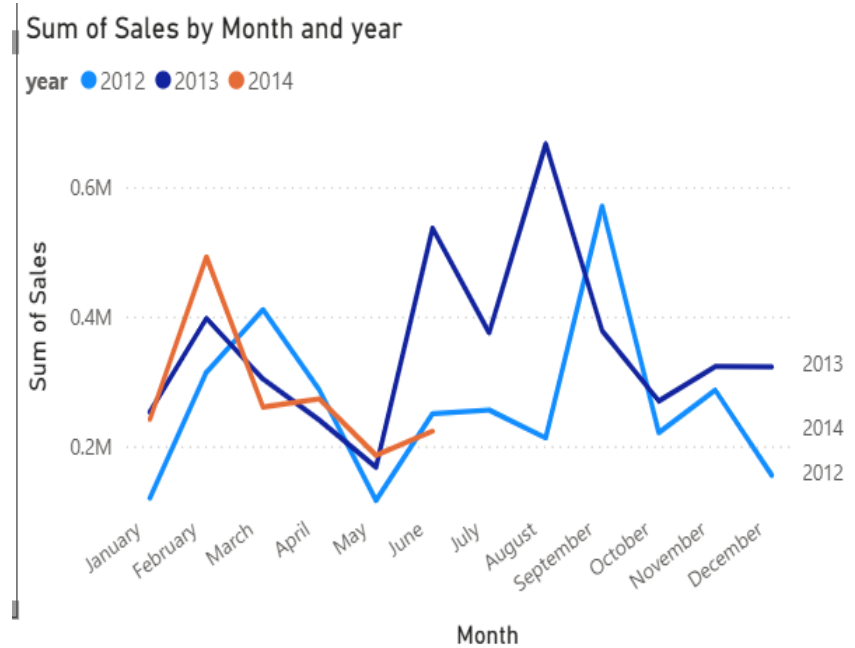
Are there any trends in sales between the two regions? Is there any seasonality?

USA



Both the U.S. and Europe show seasonal sales patterns, with noticeable peaks in **February, March, August and September** across the years. The U.S. trends are more stable, while Europe shows more ups and downs, especially in 2013.

Europe



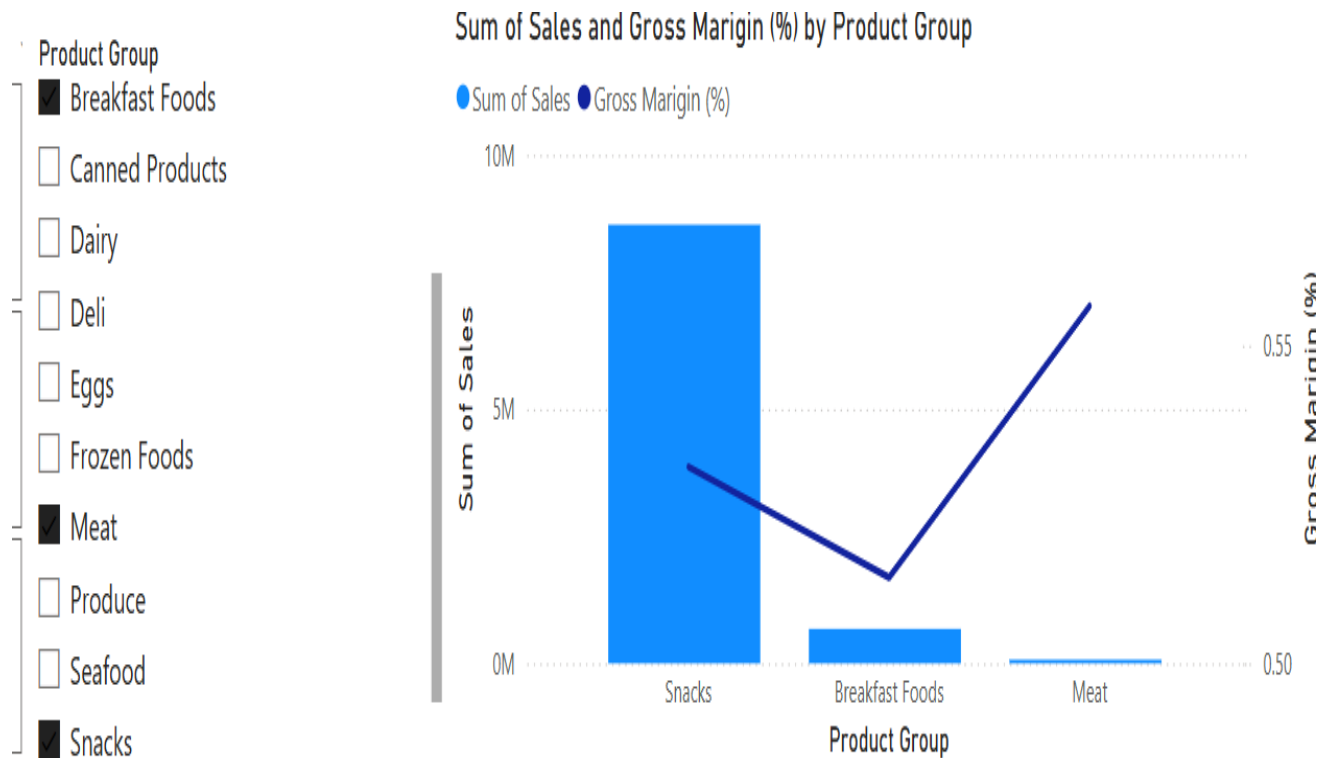
Question 4:

Compare the sales and margins of the following product groups over the years 2012, 2013, and 2014 YTD:

a. Breakfast Foods

b. Meat

c. Snacks



Among the three product groups, Snacks had the highest sales but only a medium profit margin.

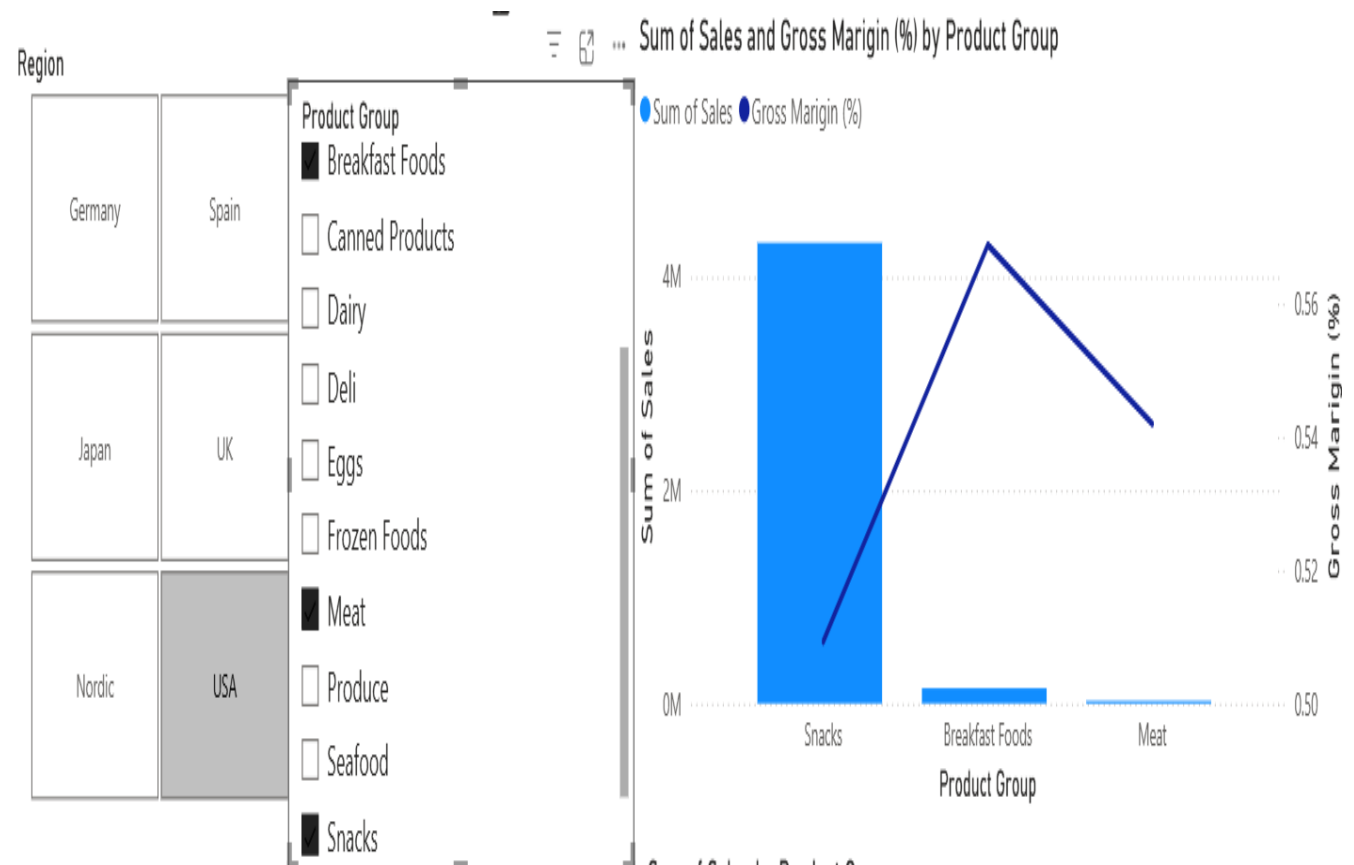
Meat had the lowest sales but delivered the highest gross margin percentage, making it the most profitable per unit sold.

Breakfast Foods showed low performance in both sales and profitability compared to the other two.

Question 5:

How do the trends differ when you look at them separately for the U.S. and European regions?

For USA

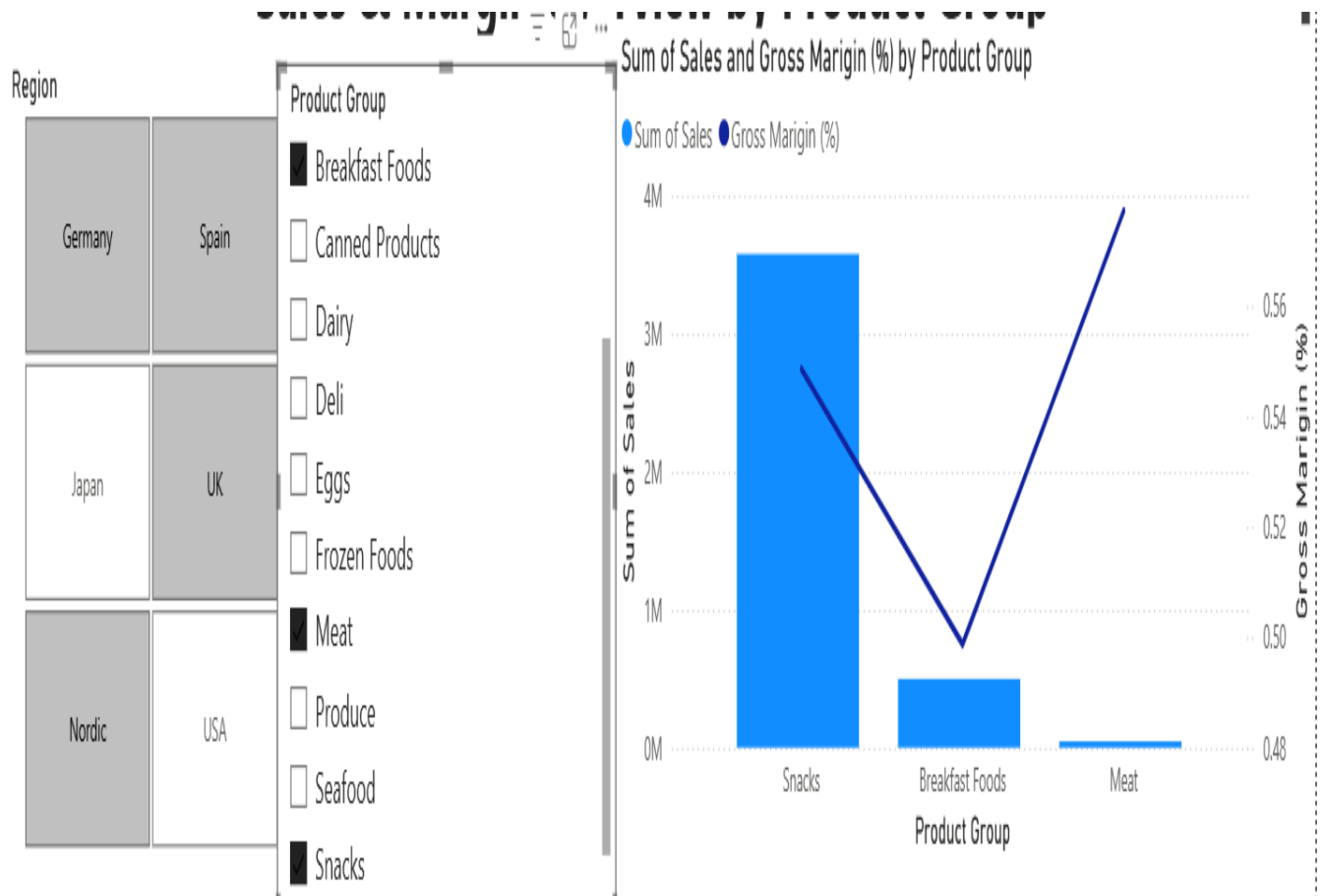


Regionally, Snacks performed best, with the USA contributing the highest sales overall.

Meat and Breakfast Foods had lower sales across all regions, with Meat showing especially low volume despite its higher margin.

This suggests that Snacks are popular and widely sold across multiple regions, while Meat may be a high-margin niche product.

For Europe



In Europe, Snacks generated the highest sales among the three product groups, while Meat had the lowest sales but the highest gross margin percentage. Breakfast Foods had moderate sales, but the lowest profitability. This shows that while Snacks are the top-selling product group in Europe, Meat offers greater profitability per unit, making it a strategic focus for margin improvement.

Question 6:

Based on these changes, is there a unified strategy we might recommend to maximize portfolio profitability to the U.S. and Europe?

To improve profitability, the company should focus on **increasing the sales of Meat**, which has the **highest gross margin** despite low sales.

At the same time, it should explore ways to **increase the margin on Snacks**, since they already generate the highest sales volume.

Promoting high-margin items and improving pricing or cost-efficiency in high-volume categories will help maximize profit.

Question 7:

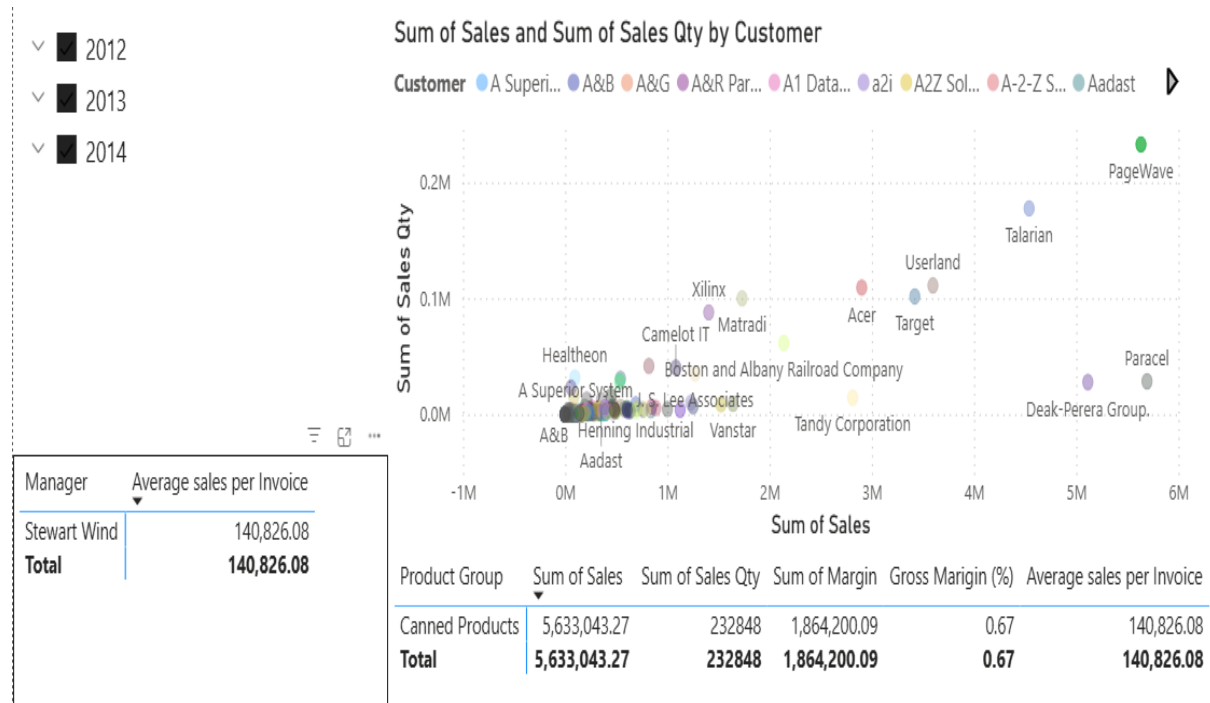
Which manager's customers produce the highest average sales per invoice?

Manager	Average sales per Invoice
Dennis Johnson	11,562.75
Stewart Wind	7,396.86
Carolyn Halmon	6,209.52
Micheal Williams	4,197.67
John Greg	4,049.28
Kathy Clinton	2,833.95
David Laychak	2,756.54
Brenda Gibson	2,538.13
Amanda Honda	1,979.28
William Fisher	1,684.91
Molly McKenzie	1,574.52
Stephanie Reagan	1,549.72
John Davis	1,533.00
Sheila Hein	1,297.66
Samantha Allen	1,182.52
Odessa Morris	1,005.39
Ken Roberts	894.76
Viginia Mountain	797.72
Total	2,736.67

The manager whose customers produce the highest average sales per invoice is Dennis Johnson, with an average of \$11,562.75 per invoice.

Question 8:

Who are the customers that are generating both the highest volume and highest value sales?



The customer generating both the highest sales volume and the highest sales value is PageWave.

With over \$5.63 million in sales and more than 232,000 units sold, PageWave is the company's most valuable customer in terms of both quantity and revenue.

Question 9:

How do the margins on these customers' sales compare to other accounts?

Customer	Sum of Sales	Sum of Sales Qty	Sum of Margin	Gross Margin (%)	Average sales per Invoice
Paracel	5,692,255.83	28433	2,610,382.11	0.54	37,204.29
Deak-Perera Group.	5,113,517.05	27789	2,216,321.14	0.57	37,877.90
PageWave	5,633,043.27	232848	1,864,200.09	0.67	140,826.08
Talarian	4,538,862.73	177641	1,646,496.04	0.64	85,638.92
Tandy Corporation	2,812,720.85	14334	1,348,829.24	0.52	37,502.94
Target	3,420,475.85	101631	1,113,013.44	0.67	100,602.23
Acer	2,901,027.96	109296	1,108,315.89	0.62	170,648.70
Userland	3,597,542.40	111200	1,076,405.60	0.70	449,692.80
Boston and Albany Railroad Company	2,138,197.60	61600	740,163.20	0.65	534,549.40
Vanstar	1,641,594.45	9068	706,635.68	0.57	1,217.80
Teammax	1,250,856.29	7062	659,510.42	0.47	13,745.67
Kerite Company	1,523,686.76	8582	622,921.10	0.59	1,378.90
Unitec Systems	1,227,287.46	10040	574,923.59	0.53	64,594.08

PageWave has a gross margin of 67%, which is higher than most other customers and the same as Target.

While Userland has the highest margin at 70%, PageWave is still very profitable. This means PageWave is not only a top buyer but also a strong source of profit.

Question 10:

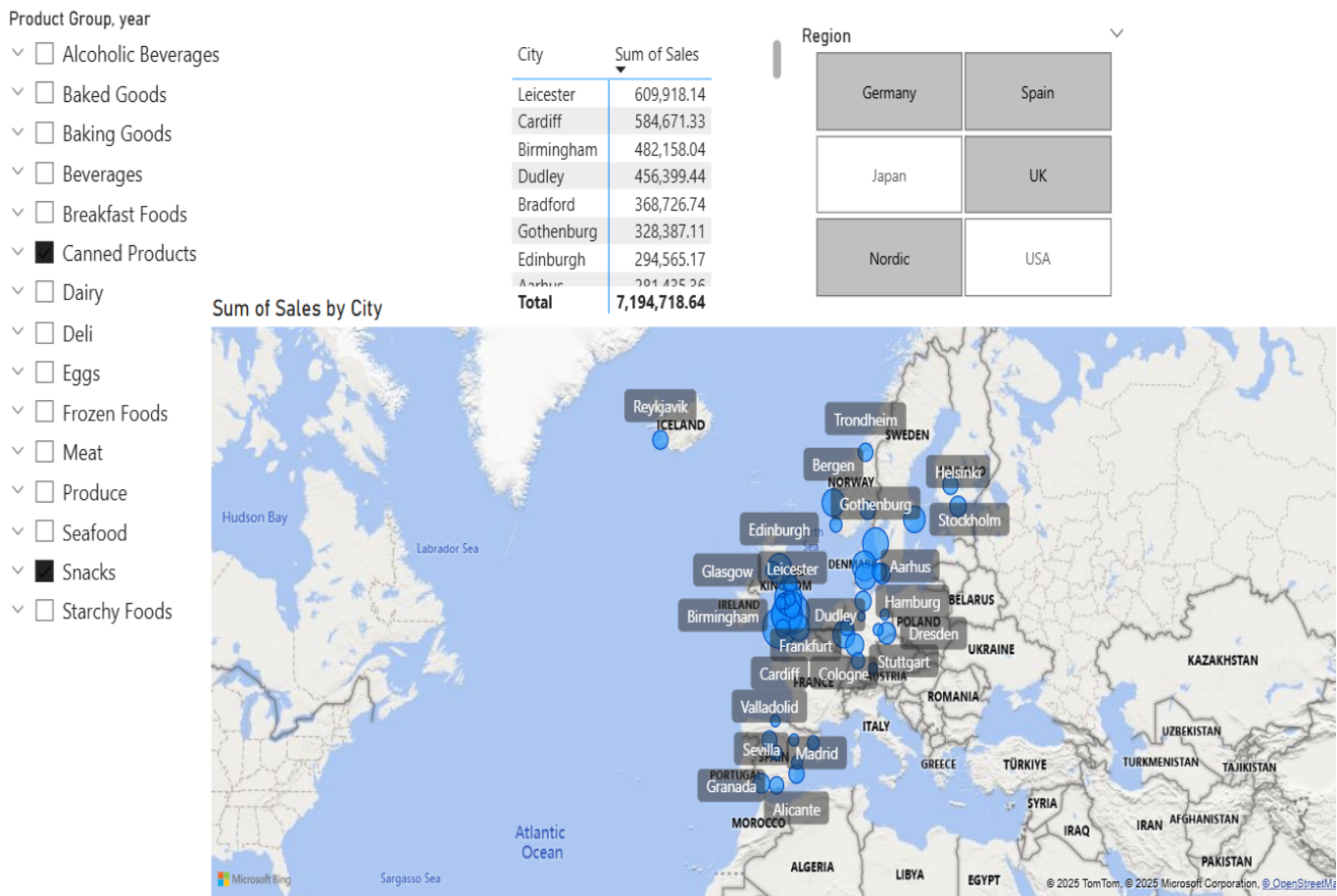
What are the most frequent product types bought by these customers?

Product Group	Sum of Sales	Sum of Sales Qty	Sum of Margin	Gross Margin (%)	Average sales per Invoice
Produce	24,155,515.79	434930	9,450,011.10	0.61	2,158.67
Canned Products	20,520,053.74	608595	7,718,976.66	0.62	2,593.86
Deli	14,634,768.73	143445	6,163,060.97	0.58	2,450.56
Frozen Foods	9,485,973.64	90592	4,641,658.28	0.51	1,179.41
Snacks	8,626,830.19	93352	4,046,628.63	0.53	910.10
Dairy	7,180,056.96	110773	2,354,666.55	0.67	1,142.59
Baking Goods	6,732,171.36	47183	3,220,045.96	0.52	1,149.03
Beverages	6,323,149.53	103482	2,733,164.29	0.57	1,523.65
Starchy Foods	3,435,909.19	19600	1,659,479.09	0.52	1,441.84
Alcoholic Beverages	2,278,029.67	142638	521,773.66	0.77	914.87
Breakfast Foods	678,251.12	6189	329,948.79	0.51	521.73
Baked Goods	442,300.89	9681	230,113.86	0.48	169.40
Eggs	245,221.95	4599	125,591.69	0.49	169.59
Meat	72,541.16	677	32,191.19	0.56	320.98
Seafood	41,900.89	636	25,878.39	0.38	103.46
Total	104,852,674.81	1816372	43,253,189.11	0.59	2,736.67

The most popular product groups among top customers are Produce, Canned Products, and Deli.

Question 11:

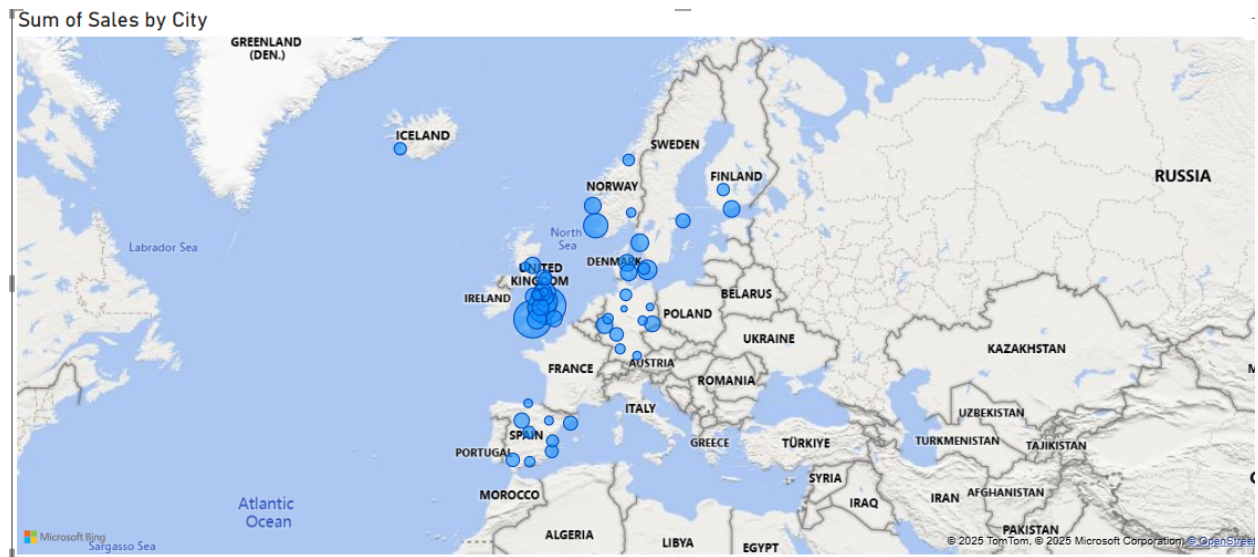
Are there European cities that appear to be dominating sales of the canned goods and snacks product groups?



The top cities in Europe for sales are Leicester, Cardiff, and Birmingham. These cities have the highest sales in Europe, especially in product groups like Canned Products and Snacks.

Question 12:

What countries could we focus on to help increase our presence in Europe?



Region	Sum of Sales	Gross Margin (%)	City	Sum of Sales	Gross Margin (%)
UK	28,157,182.23	0.55	Cardiff	6,284,806.41	0.54
Nordic	10,382,965.91	0.59	Leicester	6,177,789.74	0.57
Spain	3,449,601.72	0.54	Birmingham	3,263,878.04	0.52
Germany	3,323,936.92	0.56	Stavanger	2,275,834.03	0.67
Total	45,313,686.78	0.56	Wakefield	1,899,765.78	0.52
			Nottingham	1,603,137.28	0.72
			Malmö	1,335,637.21	0.64
			Driftal	1,204,501.65	0.40
			Total	45,313,686.78	0.56

To grow our presence in Europe, we should focus on Germany, Spain, and the Nordic countries.

These countries are already part of our market but show lower sales activity compared to the UK which can make it a strong potential for future expansion. Nordic cities like Stavanger and Malmö, though smaller in sales, have high profit margins, making them strong candidates for growth-focused campaigns.

